Building quantum codes with transversal CCZ

from projective geometry and SAT solvers

Bohan Lu, Duke University

Joint work with Ken Brown

Virginia Tech Applied Algebra Seminar

October 22, 2026

arXiv:2610.09341

Outline

1. **Quantum error correction and transversal gates**
2. The cube code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

Pauli strings and stabilizer codes

The state of n qubits lies in $(\mathbb{C}^2)^{\otimes n}$, with basis $|c\rangle$ for $c \in \mathbb{F}_2^n$. For binary vectors $c, c' \in \mathbb{F}_2^n$, the weight $|c|$ is the number of ones of c, $c \oplus c'$ is the entrywise sum mod 2 (XOR, not a direct sum), the *overlap* $c \wedge c'$ is the entrywise product, and $c \cdot c' = \sum_i c_i c'_i \bmod 2 = |c \wedge c'| \bmod 2$. For $a, b \in \mathbb{F}_2^n$, the *Pauli strings* $X(a) = X^{a_1} \otimes \cdots \otimes X^{a_n}$ and $Z(b) = Z^{b_1} \otimes \cdots \otimes Z^{b_n}$ act by $X(a)|c\rangle = |c \oplus a\rangle$ and $Z(b)|c\rangle = (-1)^{b \cdot c}|c\rangle$.

Pauli strings and stabilizer codes

The state of n qubits lies in $(\mathbb{C}^2)^{\otimes n}$, with basis $|c\rangle$ for $c \in \mathbb{F}_2^n$. For binary vectors $c, c' \in \mathbb{F}_2^n$, the weight $|c|$ is the number of ones of c, $c \oplus c'$ is the entrywise sum mod 2 (XOR, not a direct sum), the *overlap* $c \wedge c'$ is the entrywise product, and $c \cdot c' = \sum_i c_i c'_i \bmod 2 = |c \wedge c'| \bmod 2$. For $a, b \in \mathbb{F}_2^n$, the *Pauli strings* $X(a) = X^{a_1} \otimes \cdots \otimes X^{a_n}$ and $Z(b) = Z^{b_1} \otimes \cdots \otimes Z^{b_n}$ act by $X(a)|c\rangle = |c \oplus a\rangle$ and $Z(b)|c\rangle = (-1)^{b \cdot c}|c\rangle$.

- Pauli strings commute or anticommute: $X(a)Z(b) = (-1)^{a \cdot b} Z(b)X(a)$.

Pauli strings and stabilizer codes

The state of n qubits lies in $(\mathbb{C}^2)^{\otimes n}$, with basis $|c\rangle$ for $c \in \mathbb{F}_2^n$. For binary vectors $c, c' \in \mathbb{F}_2^n$, the weight $|c|$ is the number of ones of c, $c \oplus c'$ is the entrywise sum mod 2 (XOR, not a direct sum), the *overlap* $c \wedge c'$ is the entrywise product, and $c \cdot c' = \sum_i c_i c'_i \bmod 2 = |c \wedge c'| \bmod 2$. For $a, b \in \mathbb{F}_2^n$, the *Pauli strings* $X(a) = X^{a_1} \otimes \cdots \otimes X^{a_n}$ and $Z(b) = Z^{b_1} \otimes \cdots \otimes Z^{b_n}$ act by $X(a)|c\rangle = |c \oplus a\rangle$ and $Z(b)|c\rangle = (-1)^{b \cdot c}|c\rangle$.

- Pauli strings commute or anticommute: $X(a)Z(b) = (-1)^{a \cdot b} Z(b)X(a)$.
- A *stabilizer code* is the joint $+1$ eigenspace of an abelian group of Pauli strings, the stabilizers, not containing $-I$. An error, modeled as a Pauli string, that anticommutes with a stabilizer flips its measured eigenvalue. These eigenvalues form the *syndrome*.

Pauli strings and stabilizer codes

The state of n qubits lies in $(\mathbb{C}^2)^{\otimes n}$, with basis $|c\rangle$ for $c \in \mathbb{F}_2^n$. For binary vectors $c, c' \in \mathbb{F}_2^n$, the weight $|c|$ is the number of ones of c, $c \oplus c'$ is the entrywise sum mod 2 (XOR, not a direct sum), the *overlap* $c \wedge c'$ is the entrywise product, and $c \cdot c' = \sum_i c_i c'_i \bmod 2 = |c \wedge c'| \bmod 2$. For $a, b \in \mathbb{F}_2^n$, the *Pauli strings* $X(a) = X^{a_1} \otimes \cdots \otimes X^{a_n}$ and $Z(b) = Z^{b_1} \otimes \cdots \otimes Z^{b_n}$ act by $X(a)|c\rangle = |c \oplus a\rangle$ and $Z(b)|c\rangle = (-1)^{b \cdot c}|c\rangle$.

- Pauli strings commute or anticommute: $X(a)Z(b) = (-1)^{a \cdot b} Z(b)X(a)$.
- A *stabilizer code* is the joint $+1$ eigenspace of an abelian group of Pauli strings, the stabilizers, not containing $-I$. An error, modeled as a Pauli string, that anticommutes with a stabilizer flips its measured eigenvalue. These eigenvalues form the *syndrome*.
- The weight of $X(a)Z(b)$ is the number of qubits where a or b is 1. An $[\![n, k, d]\!]$ code has dimension 2^k, and d is the minimum weight of a Pauli string that commutes with every stabilizer but is not a stabilizer up to sign (a nontrivial *logical operator*).

CSS codes

For binary linear codes $\mathcal{C}_2 \subseteq \mathcal{C}_1 \subseteq \mathbb{F}_2^n$, the CSS code [CS96, Ste96] is the stabilizer code with X-*type* stabilizers $X(a)$, $a \in \mathcal{C}_2$, and Z-*type* stabilizers $Z(b)$, $b \in \mathcal{C}_1^\perp$. They commute because $a \cdot b = 0$. Write $\mathcal{C}_2 = \operatorname{rowspan} S$ and $\mathcal{C}_1 = \operatorname{rowspan} G$, $G = \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right]$, with stabilizer rows S_l of S ($s \times n$), so the $X(S_l)$ generate the X-type stabilizers, and logical rows K_i of K ($k \times n$). The logical basis, a superposition over the coset $K^\top x \oplus \mathcal{C}_2$, is

$$|x\rangle_L = 2^{-s/2} \sum_{y \in \mathbb{F}_2^s} |K^\top x \oplus S^\top y\rangle, \qquad x \in \mathbb{F}_2^k.$$

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

CSS codes

For binary linear codes $\mathcal{C}_2 \subseteq \mathcal{C}_1 \subseteq \mathbb{F}_2^n$, the CSS code [CS96, Ste96] is the stabilizer code with X-*type* stabilizers $X(a)$, $a \in \mathcal{C}_2$, and Z-*type* stabilizers $Z(b)$, $b \in \mathcal{C}_1^\perp$. They commute because $a \cdot b = 0$. Write $\mathcal{C}_2 = \operatorname{rowspan} S$ and $\mathcal{C}_1 = \operatorname{rowspan} G$, $G = \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right]$, with stabilizer rows S_l of S ($s \times n$), so the $X(S_l)$ generate the X-type stabilizers, and logical rows K_i of K ($k \times n$). The logical basis, a superposition over the coset $K^\top x \oplus \mathcal{C}_2$, is

$$|x\rangle_L = 2^{-s/2} \sum_{y \in \mathbb{F}_2^s} |K^\top x \oplus S^\top y\rangle, \qquad x \in \mathbb{F}_2^k.$$

- $Z(e)$ has syndrome Se and is undetected if and only if $e \in \mathcal{C}_2^\perp$. $X(e)$ is undetected if and only if $e \in \mathcal{C}_1$. So Z errors are corrected by decoding $\mathcal{C}_2^\perp$, and X errors by decoding $\mathcal{C}_1$.

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

CSS codes

For binary linear codes $\mathcal{C}_2 \subseteq \mathcal{C}_1 \subseteq \mathbb{F}_2^n$, the CSS code [CS96, Ste96] is the stabilizer code with X-*type* stabilizers $X(a)$, $a \in \mathcal{C}_2$, and Z-*type* stabilizers $Z(b)$, $b \in \mathcal{C}_1^\perp$. They commute because $a \cdot b = 0$. Write $\mathcal{C}_2 = \operatorname{rowspan} S$ and $\mathcal{C}_1 = \operatorname{rowspan} G$, $G = \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right]$, with stabilizer rows S_l of S ($s \times n$), so the $X(S_l)$ generate the X-type stabilizers, and logical rows K_i of K ($k \times n$). The logical basis, a superposition over the coset $K^\top x \oplus \mathcal{C}_2$, is

$$|x\rangle_L = 2^{-s/2} \textstyle\sum_{y \in \mathbb{F}_2^s} |K^\top x \oplus S^\top y\rangle, \qquad x \in \mathbb{F}_2^k.$$

- $Z(e)$ has syndrome Se and is undetected if and only if $e \in \mathcal{C}_2^\perp$. $X(e)$ is undetected if and only if $e \in \mathcal{C}_1$. So Z errors are corrected by decoding $\mathcal{C}_2^\perp$, and X errors by decoding $\mathcal{C}_1$.
- Logical operators: $X(a)$, $a \in \mathcal{C}_1 \setminus \mathcal{C}_2$, and $Z(b)$, $b \in \mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp$. Their minimum weights are d_X and d_Z, and the code is $[\![n, k, d]\!]$ with $k = \dim \mathcal{C}_1 - \dim \mathcal{C}_2$, $d = \min(d_X, d_Z)$.

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

Clifford and non-Clifford gates

On the basis states, with $x, x_1, x_2, x_3 \in \mathbb{F}_2$:

$$H|x\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle + (-1)^x|1\rangle\big), \quad S|x\rangle = i^x|x\rangle, \quad \mathrm{CNOT}|x_1x_2\rangle = |x_1,\, x_1 \oplus x_2\rangle,$$
$$T|x\rangle = e^{i\pi x/4}|x\rangle, \quad \mathrm{CCZ}|x_1x_2x_3\rangle = (-1)^{x_1x_2x_3}|x_1x_2x_3\rangle.$$

S, T and CCZ are *diagonal*: each multiplies $|x\rangle$ by a phase.

Clifford and non-Clifford gates

On the basis states, with $x, x_1, x_2, x_3 \in \mathbb{F}_2$:

$$\text{Clifford:} \quad H|x\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle + (-1)^x|1\rangle\big), \quad S|x\rangle = i^x|x\rangle, \quad \mathrm{CNOT}|x_1x_2\rangle = |x_1,\, x_1 \oplus x_2\rangle,$$

$$\text{non-Clifford:} \quad T|x\rangle = e^{i\pi x/4}|x\rangle, \quad \mathrm{CCZ}|x_1x_2x_3\rangle = (-1)^{x_1x_2x_3}|x_1x_2x_3\rangle.$$

S, T and CCZ are *diagonal*: each multiplies $|x\rangle$ by a phase.

- The *Clifford group* is the group of unitaries that map Pauli strings to Pauli strings under conjugation, up to phase. H, S and CNOT generate it. T and CCZ are not Clifford.

Clifford and non-Clifford gates

On the basis states, with $x, x_1, x_2, x_3 \in \mathbb{F}_2$:

$$\text{Clifford:} \quad H|x\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle + (-1)^x|1\rangle\big), \quad S|x\rangle = i^x|x\rangle, \quad \mathrm{CNOT}|x_1x_2\rangle = |x_1,\, x_1 \oplus x_2\rangle,$$

$$\text{non-Clifford:} \quad T|x\rangle = e^{i\pi x/4}|x\rangle, \quad \mathrm{CCZ}|x_1x_2x_3\rangle = (-1)^{x_1x_2x_3}|x_1x_2x_3\rangle.$$

S, T and CCZ are *diagonal*: each multiplies $|x\rangle$ by a phase.

- The *Clifford group* is the group of unitaries that map Pauli strings to Pauli strings under conjugation, up to phase. H, S and CNOT generate it. T and CCZ are not Clifford.
- A gate set is *universal* if it approximates every unitary. Clifford gates alone are not universal. Adding T or CCZ gives a universal gate set. So an error-corrected quantum computer must apply Clifford gates and at least one non-Clifford gate to the encoded qubits.

Transversal gates and magic-state distillation

A *logical* U is a unitary $\bar{U}$ that acts on $|x\rangle_L$ as U acts on $|x\rangle$. It is *single-qubit transversal* if $\bar{U} = U_1 \otimes \cdots \otimes U_n$ on the code space, with U_j acting on qubit j. If one U_j fails, the error stays on qubit j.

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.

[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Transversal gates and magic-state distillation

A *logical* U is a unitary $\bar{U}$ that acts on $|x\rangle_L$ as U acts on $|x\rangle$. It is *single-qubit transversal* if $\bar{U} = U_1 \otimes \cdots \otimes U_n$ on the code space, with U_j acting on qubit j. If one U_j fails, the error stays on qubit j.

- Eastin–Knill [EK09]: no code that detects every single-qubit error has a universal set of transversal logical gates. At least one logical gate must be implemented another way.

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.
[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Transversal gates and magic-state distillation

A *logical* U is a unitary $\bar{U}$ that acts on $|x\rangle_L$ as U acts on $|x\rangle$. It is *single-qubit transversal* if $\bar{U} = U_1 \otimes \cdots \otimes U_n$ on the code space, with U_j acting on qubit j. If one U_j fails, the error stays on qubit j.

- Eastin–Knill [EK09]: no code that detects every single-qubit error has a universal set of transversal logical gates. At least one logical gate must be implemented another way.
- One mainstream way is magic-state distillation [BK05]. Clifford operations turn many noisy copies of a non-Clifford state, such as $T|+\rangle$ or $\mathrm{CCZ}|{+}{+}{+}\rangle$ with $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, into fewer copies with lower error. Each copy is consumed to apply one logical non-Clifford gate.

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.
[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Transversal gates and magic-state distillation

A *logical* U is a unitary $\bar{U}$ that acts on $|x\rangle_L$ as U acts on $|x\rangle$. It is *single-qubit transversal* if $\bar{U} = U_1 \otimes \cdots \otimes U_n$ on the code space, with U_j acting on qubit j. If one U_j fails, the error stays on qubit j.

- Eastin–Knill [EK09]: no code that detects every single-qubit error has a universal set of transversal logical gates. At least one logical gate must be implemented another way.
- One mainstream way is magic-state distillation [BK05]. Clifford operations turn many noisy copies of a non-Clifford state, such as $T|+\rangle$ or $\mathrm{CCZ}|{+}{+}{+}\rangle$ with $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, into fewer copies with lower error. Each copy is consumed to apply one logical non-Clifford gate.
- Most distillation protocols are built from a stabilizer code on which a single-qubit transversal gate acts as a logical non-Clifford gate, such as T or CCZ. This talk finds such codes.

[EK09] B. Eastin and E. Knill, Restrictions on transversal encoded quantum gate sets, Phys. Rev. Lett. 102, 110502 (2009), arXiv:0811.4262.
[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

Codes with different transversal gates

Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode	$T^{\otimes 15}$ is logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	T or $T^\dagger$ on each qubit: logical CCZ.

$\mathrm{RM}(r,m)$: Reed–Muller code, degree $\leq r$ in m variables. Punctured: one coordinate deleted. $\mathbf{1}$: all-ones vector.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Codes with different transversal gates

Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode	$T^{\otimes 15}$ is logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	T or $T^\dagger$ on each qubit: logical CCZ.

$\mathrm{RM}(r,m)$: Reed–Muller code, degree $\leq r$ in m variables. Punctured: one coordinate deleted. $\mathbf{1}$: all-ones vector.

CNOT between two code blocks is transversal for every CSS code. Since Clifford $+T$ is universal, [EK09] rules out transversal T for Steane and transversal H for $[\![15,1,3]\!]$.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Codes with different transversal gates

Code	$\mathcal{C}_1 \supset \mathcal{C}_2$	Transversal logical gates
Steane $[\![7,1,3]\!]$ [Ste96]	Hamming $[7,4,3] \supset$ its dual $[7,3,4]$	H, S. Not T.
$[\![15,1,3]\!]$ [BK05]	punctured $\mathrm{RM}(1,4)$, $[15,5,7] \supset$ its even-weight subcode	$T^{\otimes 15}$ is logical $T^\dagger$. Not H.
$[\![8,3,2]\!]$ [CH17]	$\mathrm{RM}(1,3) \supset \langle \mathbf{1} \rangle$	T or $T^\dagger$ on each qubit: logical CCZ.

$\mathrm{RM}(r,m)$: Reed–Muller code, degree $\leq r$ in m variables. Punctured: one coordinate deleted. $\mathbf{1}$: all-ones vector.

CNOT between two code blocks is transversal for every CSS code. Since Clifford $+T$ is universal, [EK09] rules out transversal T for Steane and transversal H for $[\![15,1,3]\!]$.

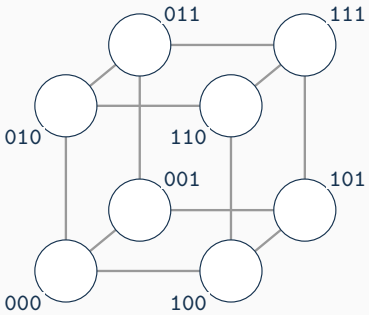
Question: which CSS codes with $k=3$ have $U_j \in \{T, T^\dagger\}$ with $U_1 \otimes \cdots \otimes U_n$ logical CCZ?

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Outline

1. Quantum error correction and transversal gates
2. **The cube code: a worked example of logical CCZ**
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

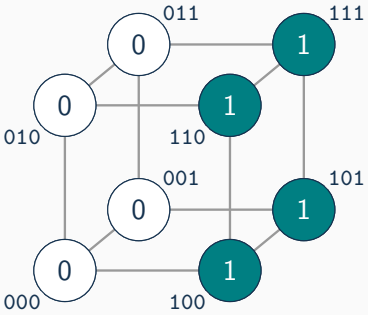
Qubits indexed by the points of $\mathbb{F}_2^3$



Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$. Its weight $|c|$ is the number of points where $c = 1$, and the overlap $c \wedge c'$ is 1 exactly on the intersection of the supports of c and c'.

Qubits indexed by the points of $\mathbb{F}_2^3$



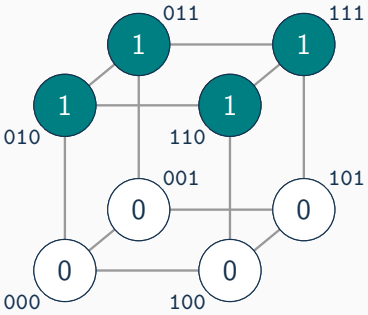
The vector u_1.

Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$. Its weight $|c|$ is the number of points where $c = 1$, and the overlap $c \wedge c'$ is 1 exactly on the intersection of the supports of c and c'.

Each polynomial in $u_1, \ldots, u_m$, evaluated at the 2^m points, gives a binary vector. For example, u_1 gives 00001111 in the order $000, 001, \ldots, 111$.

Qubits indexed by the points of $\mathbb{F}_2^3$



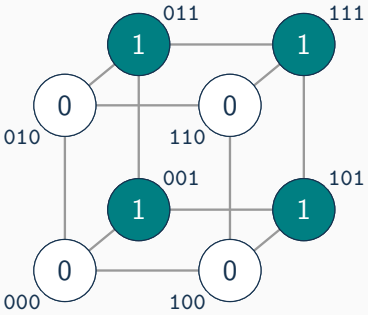
The vector u_2.

Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$. Its weight $|c|$ is the number of points where $c = 1$, and the overlap $c \wedge c'$ is 1 exactly on the intersection of the supports of c and c'.

Each polynomial in $u_1, \ldots, u_m$, evaluated at the 2^m points, gives a binary vector. For example, u_1 gives 00001111 in the order $000, 001, \ldots, 111$.

Qubits indexed by the points of $\mathbb{F}_2^3$



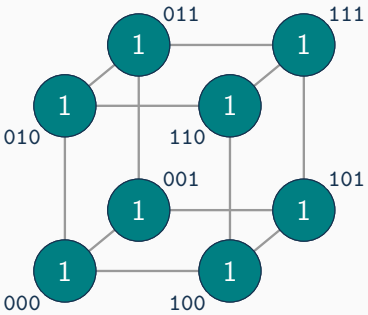
The vector u_3.

Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$. Its weight $|c|$ is the number of points where $c = 1$, and the overlap $c \wedge c'$ is 1 exactly on the intersection of the supports of c and c'.

Each polynomial in $u_1, \ldots, u_m$, evaluated at the 2^m points, gives a binary vector. For example, u_1 gives 00001111 in the order $000, 001, \ldots, 111$.

Qubits indexed by the points of $\mathbb{F}_2^3$



The constant vector $\mathbf{1}$.

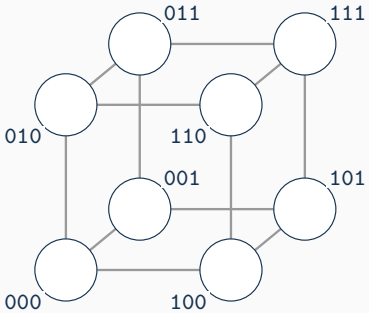
Index the $n = 2^m$ qubits by the points $u \in \mathbb{F}_2^m$. Here $m = 3$ and $n = 8$.

A binary vector c of length n is then a Boolean function $c\colon \mathbb{F}_2^m \to \mathbb{F}_2$. Its weight $|c|$ is the number of points where $c = 1$, and the overlap $c \wedge c'$ is 1 exactly on the intersection of the supports of c and c'.

Each polynomial in $u_1, \ldots, u_m$, evaluated at the 2^m points, gives a binary vector. For example, u_1 gives 00001111 in the order $000, 001, \ldots, 111$.

The polynomials of degree at most 1 give $\mathrm{RM}(1,3) = \mathrm{span}\{\mathbf{1}, u_1, u_2, u_3\}$, an $[8, 4, 4]$ code.

The cube code $[\![8, 3, 2]\!]$

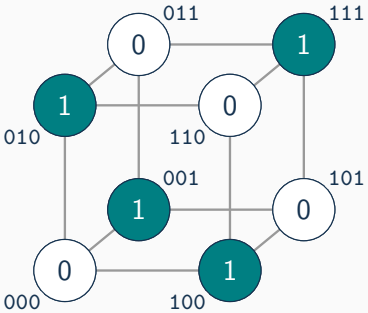


Take $S = \mathbf{1}$, so the only X-type stabilizer generator is $X(\mathbf{1}) = X^{\otimes 8}$, and let K have the rows u_1, u_2, u_3. Then $\mathcal{C}_1 = \mathrm{RM}(1, 3)$ and $k = 3$.

$K^\top x = x \cdot u = x_1 u_1 \oplus x_2 u_2 \oplus x_3 u_3$, so

$$|x\rangle_L = \big(|x \cdot u\rangle + |\mathbf{1} \oplus x \cdot u\rangle\big)/\sqrt{2}.$$

The cube code $[\![8, 3, 2]\!]$



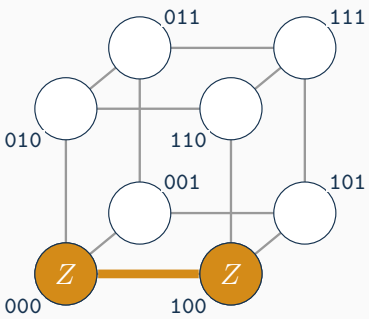
$x \cdot u = u_1 \oplus u_2 \oplus u_3$ for $x = 111$.

Take $S = \mathbf{1}$, so the only X-type stabilizer generator is $X(\mathbf{1}) = X^{\otimes 8}$, and let K have the rows u_1, u_2, u_3. Then $\mathcal{C}_1 = \mathrm{RM}(1, 3)$ and $k = 3$.

$K^\top x = x \cdot u = x_1 u_1 \oplus x_2 u_2 \oplus x_3 u_3$, so

$$|x\rangle_L = \big(|x \cdot u\rangle + |\mathbf{1} \oplus x \cdot u\rangle\big)/\sqrt{2}.$$

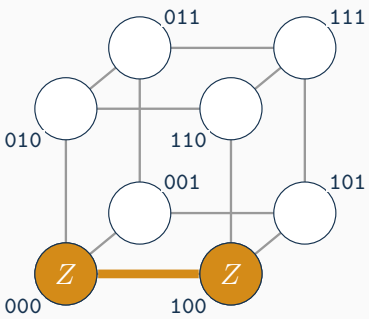
Distances of the cube code



Z on the qubits 000 and 100.

Let δ_u be the indicator of the point u. Z on the qubits 000 and 100 is $Z(e)$ with $e = \delta_{000} \oplus \delta_{100}$.

Distances of the cube code

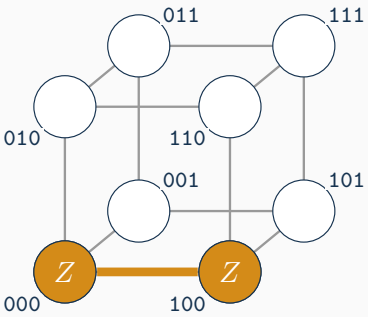


Z on the qubits 000 and 100.

Let δ_u be the indicator of the point u. Z on the qubits 000 and 100 is $Z(e)$ with $e = \delta_{000} \oplus \delta_{100}$.

- $e \cdot \mathbf{1} = 0$, so $Z(e)$ commutes with the stabilizer $X(\mathbf{1})$. Each $Z(\delta_u)$ anticommutes with it, so $d_Z \geq 2$.

Distances of the cube code

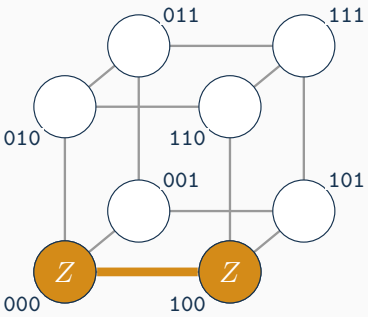


Z on the qubits 000 and 100.

Let δ_u be the indicator of the point u. Z on the qubits 000 and 100 is $Z(e)$ with $e = \delta_{000} \oplus \delta_{100}$.

- $e \cdot \mathbf{1} = 0$, so $Z(e)$ commutes with the stabilizer $X(\mathbf{1})$. Each $Z(\delta_u)$ anticommutes with it, so $d_Z \geq 2$.
- Every codeword of $\mathcal{C}_1$ is $c = x \cdot u \oplus y\mathbf{1}$ with $y \in \mathbb{F}_2$, so $c(000) = y$ and $c(100) = x_1 \oplus y$. So $Z(e)|x\rangle_L = (-1)^{x_1}|x\rangle_L$, a logical operator of weight 2, and $d_Z = 2$.

Distances of the cube code



Z on the qubits 000 and 100.

Let δ_u be the indicator of the point u. Z on the qubits 000 and 100 is $Z(e)$ with $e = \delta_{000} \oplus \delta_{100}$.

- $e \cdot \mathbf{1} = 0$, so $Z(e)$ commutes with the stabilizer $X(\mathbf{1})$. Each $Z(\delta_u)$ anticommutes with it, so $d_Z \geq 2$.
- Every codeword of $\mathcal{C}_1$ is $c = x \cdot u \oplus y\mathbf{1}$ with $y \in \mathbb{F}_2$, so $c(000) = y$ and $c(100) = x_1 \oplus y$. So $Z(e)|x\rangle_L = (-1)^{x_1}|x\rangle_L$, a logical operator of weight 2, and $d_Z = 2$.
- $\mathrm{RM}(1,3)$ has minimum weight 4, so $d_X = 4$.

Codeword weight, physical phase, and logical phase

Write $U_j = T^{\Gamma_j}$ with $\Gamma \in \{\pm 1\}^n$, where $T^{-1} = T^\dagger$. Since $T|x\rangle = e^{i\pi x/4}|x\rangle$ on each qubit,

$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi\,\mathrm{sw}_\Gamma(c)/4}|c\rangle, \qquad \mathrm{sw}_\Gamma(c) = \sum_j \Gamma_j c_j.$$

The *signed weight* $\mathrm{sw}_\Gamma(c)$ counts the ones of c on T qubits minus the ones on $T^\dagger$ qubits. We call $\bigotimes_j T^{\Gamma_j}$ a $T/T^\dagger$ *layer*.

Codeword weight, physical phase, and logical phase

Write $U_j = T^{\Gamma_j}$ with $\Gamma \in \{\pm 1\}^n$, where $T^{-1} = T^\dagger$. Since $T|x\rangle = e^{i\pi x/4}|x\rangle$ on each qubit,

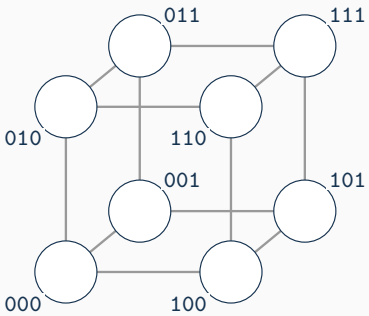
$$\bigotimes_j T^{\Gamma_j}|c\rangle = e^{i\pi\,\mathrm{sw}_\Gamma(c)/4}|c\rangle, \qquad \mathrm{sw}_\Gamma(c) = \sum_j \Gamma_j c_j.$$

The *signed weight* $\mathrm{sw}_\Gamma(c)$ counts the ones of c on T qubits minus the ones on $T^\dagger$ qubits. We call $\bigotimes_j T^{\Gamma_j}$ a $T/T^\dagger$ *layer*.

The layer acts as logical CCZ if and only if every codeword of the coset $K^\top x \oplus \mathcal{C}_2$ receives the phase $(-1)^{x_1x_2x_3}$:

$$\mathrm{sw}_\Gamma(K^\top x \oplus S^\top y) \equiv 4x_1x_2x_3 \pmod 8 \qquad \text{for all } x \in \mathbb{F}_2^3,\ y \in \mathbb{F}_2^s.$$

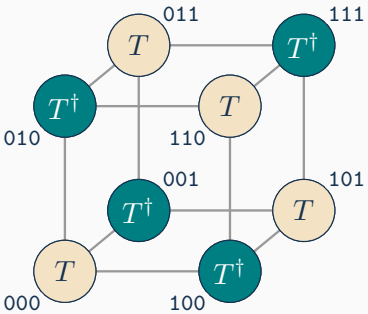
Choosing the signs on the cube



Every codeword $c = x \cdot u \oplus y\mathbf{1}$ of $\mathcal{C}_1$ is an *affine function* of u. For $x \neq 000$, the hyperplane $x \cdot u = 0$ has 4 of the 8 points, so c has exactly 4 ones.

Let $p = u_1 \oplus u_2 \oplus u_3$, the codeword $x \cdot u$ for $x = 111$.

Choosing the signs on the cube



$T^\dagger$ where $p(u) = 1$ [CH17].

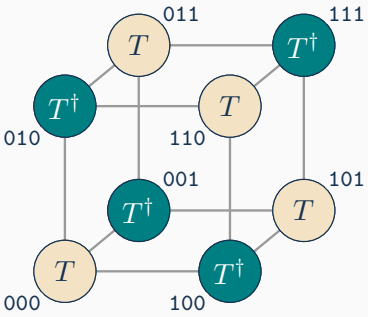
Every codeword $c = x \cdot u \oplus y\mathbf{1}$ of $\mathcal{C}_1$ is an *affine function* of u. For $x \neq 000$, the hyperplane $x \cdot u = 0$ has 4 of the 8 points, so c has exactly 4 ones.

Let $p = u_1 \oplus u_2 \oplus u_3$, the codeword $x \cdot u$ for $x = 111$.

Put $T^\dagger$ where $p = 1$, so $\Gamma_u = (-1)^{p(u)}$. Since $|p| = 4$, the signed weight depends only on the distance $|p \oplus c|$:

$$\mathrm{sw}_\Gamma(c) = |c| - 2|p \wedge c| = |p \oplus c| - 4.$$

Choosing the signs on the cube



$T^\dagger$ where $p(u) = 1$ [CH17].

Every codeword $c = x \cdot u \oplus y\mathbf{1}$ of $\mathcal{C}_1$ is an *affine function* of u. For $x \neq 000$, the hyperplane $x \cdot u = 0$ has 4 of the 8 points, so c has exactly 4 ones.

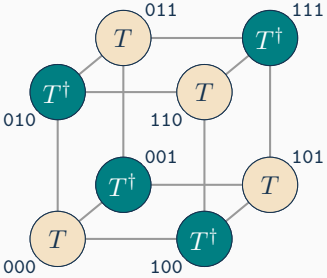
Let $p = u_1 \oplus u_2 \oplus u_3$, the codeword $x \cdot u$ for $x = 111$.

Put $T^\dagger$ where $p = 1$, so $\Gamma_u = (-1)^{p(u)}$. Since $|p| = 4$, the signed weight depends only on the distance $|p \oplus c|$:

$$\mathrm{sw}_\Gamma(c) = |c| - 2|p \wedge c| = |p \oplus c| - 4.$$

$x \neq 111$: $p \oplus c = (x \oplus 111) \cdot u \oplus y\mathbf{1}$ is nonconstant, so $|p \oplus c| = 4$ and $\mathrm{sw}_\Gamma(c) = 0$.
$x = 111$: $c \in \{p, \mathbf{1} \oplus p\}$, so $|p \oplus c| \in \{0, 8\}$ and $\mathrm{sw}_\Gamma(c) = \pm 4$, phase -1.

Reading the cube phase from overlaps



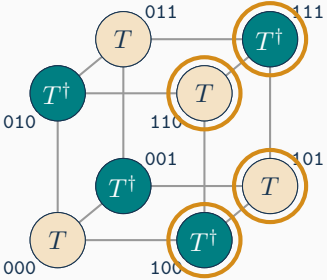
For bits a_1, a_2, a_3, XOR has the integer expansion

$$a_1 \oplus a_2 \oplus a_3 = \textstyle\sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1a_2a_3.$$

For $c = K^\top x = x_1K_1 \oplus x_2K_2 \oplus x_3K_3$, apply it at each vertex u with $a_i = x_iK_i(u)$, weight by Γ_u, and sum:

$$\begin{aligned}\mathrm{sw}_\Gamma(c) = {} & \textstyle\sum_i x_i\,\mathrm{sw}_\Gamma(K_i) - 2\sum_{i<j} x_ix_j\,\mathrm{sw}_\Gamma(K_i \wedge K_j) \\ & + 4x_1x_2x_3\ \mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3).\end{aligned}$$

Reading the cube phase from overlaps



$K_1 = u_1$: the face $u_1 = 1$.

For bits a_1, a_2, a_3, XOR has the integer expansion

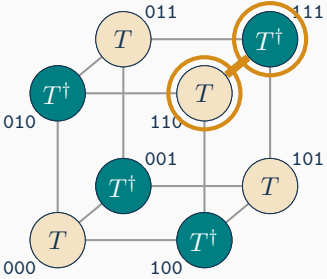
$$a_1 \oplus a_2 \oplus a_3 = \textstyle\sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1a_2a_3.$$

For $c = K^\top x = x_1K_1 \oplus x_2K_2 \oplus x_3K_3$, apply it at each vertex u with $a_i = x_iK_i(u)$, weight by Γ_u, and sum:

$$\begin{aligned}\mathrm{sw}_\Gamma(c) = {} & \textstyle\sum_i x_i\,\mathrm{sw}_\Gamma(K_i) - 2\sum_{i<j} x_ix_j\,\mathrm{sw}_\Gamma(K_i \wedge K_j) \\ & + 4x_1x_2x_3\ \mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3).\end{aligned}$$

The face $u_1 = 1$ has T at $101, 110$ and $T^\dagger$ at $100, 111$, so $\mathrm{sw}_\Gamma(K_1) = 0$. Likewise for K_2, K_3.

Reading the cube phase from overlaps



$K_1 \wedge K_2 = u_1 u_2$.

For bits a_1, a_2, a_3, XOR has the integer expansion

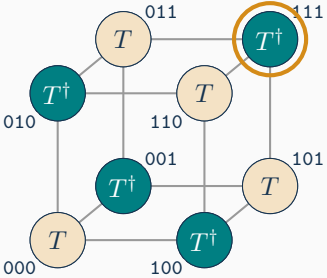
$$a_1 \oplus a_2 \oplus a_3 = \textstyle\sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1a_2a_3.$$

For $c = K^\top x = x_1K_1 \oplus x_2K_2 \oplus x_3K_3$, apply it at each vertex u with $a_i = x_iK_i(u)$, weight by Γ_u, and sum:

$$\begin{aligned}\mathrm{sw}_\Gamma(c) = {} & \textstyle\sum_i x_i\,\mathrm{sw}_\Gamma(K_i) - 2\sum_{i<j} x_ix_j\,\mathrm{sw}_\Gamma(K_i \wedge K_j) \\ & + 4x_1x_2x_3\ \mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3).\end{aligned}$$

The face $u_1 = 1$ has T at $101, 110$ and $T^\dagger$ at $100, 111$, so $\mathrm{sw}_\Gamma(K_1) = 0$. Likewise for K_2, K_3.
The edge $\{110, 111\}$ has one T and one $T^\dagger$, so $\mathrm{sw}_\Gamma(K_1 \wedge K_2) = 0$. Likewise for the other pairs.

Reading the cube phase from overlaps



$K_1 \wedge K_2 \wedge K_3 = u_1 u_2 u_3.$

For bits a_1, a_2, a_3, XOR has the integer expansion

$$a_1 \oplus a_2 \oplus a_3 = \sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1 a_2 a_3.$$

For $c = K^\top x = x_1 K_1 \oplus x_2 K_2 \oplus x_3 K_3$, apply it at each vertex u with $a_i = x_i K_i(u)$, weight by Γ_u, and sum:

$$\begin{aligned}\mathrm{sw}_\Gamma(c) = {} & \sum_i x_i \,\mathrm{sw}_\Gamma(K_i) - 2\sum_{i<j} x_i x_j \,\mathrm{sw}_\Gamma(K_i \wedge K_j) \\ & + 4x_1 x_2 x_3 \,\mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3).\end{aligned}$$

The face $u_1 = 1$ has T at $101, 110$ and $T^\dagger$ at $100, 111$, so $\mathrm{sw}_\Gamma(K_1) = 0$. Likewise for K_2, K_3.
The edge $\{110, 111\}$ has one T and one $T^\dagger$, so $\mathrm{sw}_\Gamma(K_1 \wedge K_2) = 0$. Likewise for the other pairs.
The vertex 111 is a $T^\dagger$, so $\mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3) = -1$ and $\mathrm{sw}_\Gamma(c) = -4x_1 x_2 x_3 \equiv 4x_1 x_2 x_3 \pmod 8$: logical CCZ.

The gate condition becomes conditions on generator rows

The same expansion for any number of binary vectors is the *Bravyi–Haah parity identity* [BH12]

$$\mathrm{sw}_\Gamma(a_1 \oplus \cdots \oplus a_t) = \textstyle\sum_{J \neq \emptyset} (-2)^{|J|-1}\, \mathrm{sw}_\Gamma\big(\bigwedge_{j \in J} a_j\big).$$

Expand $\mathrm{sw}_\Gamma(K^\top x \oplus S^\top y) \equiv 4x_1x_2x_3 \pmod 8$ with it and compare the coefficients of the monomials in (x, y). For distinct rows g, g', g'' of G:

$$\begin{aligned} \mathrm{sw}_\Gamma(g) &\equiv 0 \pmod 8, \\ \mathrm{sw}_\Gamma(g \wedge g') &\equiv 0 \pmod 4, \\ \mathrm{sw}_\Gamma(g \wedge g' \wedge g'') &\equiv [\{g, g', g''\} = \{K_1, K_2, K_3\}] \pmod 2. \end{aligned}$$

Here $[P]$ is 1 if the statement P holds and 0 otherwise.

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.

The gate condition becomes conditions on generator rows

The same expansion for any number of binary vectors is the *Bravyi–Haah parity identity* [BH12]

$$\mathrm{sw}_\Gamma(a_1 \oplus \cdots \oplus a_t) = \textstyle\sum_{J \neq \emptyset} (-2)^{|J|-1}\, \mathrm{sw}_\Gamma\big(\bigwedge_{j \in J} a_j\big).$$

Expand $\mathrm{sw}_\Gamma(K^\top x \oplus S^\top y) \equiv 4x_1x_2x_3 \pmod 8$ with it and compare the coefficients of the monomials in (x, y). For distinct rows g, g', g'' of G:

$$\begin{aligned} \mathrm{sw}_\Gamma(g) &\equiv 0 \pmod 8, \\ \mathrm{sw}_\Gamma(g \wedge g') &\equiv 0 \pmod 4, \\ \mathrm{sw}_\Gamma(g \wedge g' \wedge g'') &\equiv [\{g, g', g''\} = \{K_1, K_2, K_3\}] \pmod 2. \end{aligned}$$

Here $[P]$ is 1 if the statement P holds and 0 otherwise.

The $[\![8, 3, 2]\!]$ code satisfies all three: rows have $\mathrm{sw}_\Gamma = 0$, pairs of rows overlap in faces or edges with $\mathrm{sw}_\Gamma = 0$, and triples overlap in an edge ($\mathrm{sw}_\Gamma = 0$) or in the vertex 111 ($\mathrm{sw}_\Gamma = -1$).

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.

Outline

1. Quantum error correction and transversal gates
2. The cube code: a worked example of logical CCZ
3. **Triorthogonality and the CCZ conditions**
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

Historical context

2005 Bravyi–Kitaev [BK05]: magic-state distillation with ideal Clifford gates.

2012 Bravyi–Haah [BH12]: triorthogonal matrices (next slide), transversal T followed by Clifford gates.

2017 Campbell–Howard [CH17]: distillation of CCZ and other multiqubit gates.

2022 Nezami–Haah [NH22]: classify triorthogonal codes with $n + k \leq 38$ via Reed–Muller codes.

2024 Camps-Moreno, López, Matthews et al. [CLM+24]: CSS-T codes ($T^{\otimes n}$ acts as a logical gate) from cyclic and extended cyclic codes. In 2026 [CLM+26]: all diagonal transversal gates of a CSS code.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

[CLM+24] E. Camps-Moreno et al., An algebraic characterization of binary CSS-T codes and cyclic CSS-T codes for quantum fault tolerance, Quantum Inf. Process. 23, 230 (2024), arXiv:2312.17518.

[CLM+26] E. Camps-Moreno et al., Transversal gates for quantum CSS codes, arXiv:2601.21514 (2026).

Triorthogonal matrix and triorthogonal space

Matrix G [BH12]

$$|g \wedge g'| \equiv 0 \pmod 2,$$
$$|g \wedge g' \wedge g''| \equiv 0 \pmod 2.$$

Rows are distinct. Odd-weight rows give the logical X operators. Even-weight rows give the X-type stabilizers.

Space $\mathcal{H}$ [NH22]

$$\sum_{j=1}^{L} h_j h'_j h''_j \equiv 0 \pmod 2$$

for all $h, h', h'' \in \mathcal{H} \subseteq \mathbb{F}_2^L$, including repeated vectors.

Unital: $\mathbf{1}_L \in \mathcal{H}$. Every vector in $\mathcal{H}$ has even weight.

Nezami–Haah classify the spaces $\mathcal{H}$, then obtain codes by deleting (puncturing) coordinates. This work fixes S from geometry first, then searches for a compatible K.

Conditions for quasi-transversal CCZ

K1	$\lvert K_i\rvert \equiv 0$
K2	$\lvert K_i \wedge K_{i'}\rvert \equiv 0$
K3	$\lvert K_1 \wedge K_2 \wedge K_3\rvert \equiv 1$
KS1	$\lvert K_i \wedge S_l\rvert \equiv 0$
KS2	$\lvert K_i \wedge K_{i'} \wedge S_l\rvert \equiv 0$
KS3	$\lvert K_i \wedge S_l \wedge S_{l'}\rvert \equiv 0$
S1	$\lvert S_l\rvert \equiv 0$
S2	$\lvert S_l \wedge S_{l'}\rvert \equiv 0$
S3	$\lvert S_l \wedge S_{l'} \wedge S_{l''}\rvert \equiv 0$

Rows K_i of K, S_l of S. All mod 2, indices distinct.

Reduce the three row congruences mod 2. Since Γ_j is odd, $\mathrm{sw}_\Gamma(c) \equiv |c| \pmod 2$, and the result is these nine parities, the Campbell–Howard (CH) conditions [CH17]. They make $C\,T^{\otimes n}$ logical CCZ for some diagonal Clifford gate C, the *correction* (*quasi-transversal* CCZ).

Conditions for quasi-transversal CCZ

K1	$\lvert K_i\rvert \equiv 0$
K2	$\lvert K_i \wedge K_{i'}\rvert \equiv 0$
K3	$\lvert K_1 \wedge K_2 \wedge K_3\rvert \equiv 1$
KS1	$\lvert K_i \wedge S_l\rvert \equiv 0$
KS2	$\lvert K_i \wedge K_{i'} \wedge S_l\rvert \equiv 0$
KS3	$\lvert K_i \wedge S_l \wedge S_{l'}\rvert \equiv 0$
S1	$\lvert S_l\rvert \equiv 0$
S2	$\lvert S_l \wedge S_{l'}\rvert \equiv 0$
S3	$\lvert S_l \wedge S_{l'} \wedge S_{l''}\rvert \equiv 0$

Rows K_i of K, S_l of S. All mod 2, indices distinct.

Reduce the three row congruences mod 2. Since Γ_j is odd, $\mathrm{sw}_\Gamma(c) \equiv |c| \pmod 2$, and the result is these nine parities, the Campbell–Howard (CH) conditions [CH17]. They make $C\,T^{\otimes n}$ logical CCZ for some diagonal Clifford gate C, the *correction* (*quasi-transversal* CCZ).

A $T/T^\dagger$ layer with no correction needs the signed congruences mod 8 and mod 4. These are conditions on Γ as well as on G.

Outline

1. Quantum error correction and transversal gates
2. The cube code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. **Projective geometry and the 48-qubit code**
5. Column indicators, logical rows, and phase signs
6. Lower bounds and open questions

Result: a 48-qubit code with a transversal $T/T^\dagger$ CCZ

The construction [LB26]

$\mathcal{Q}_{48} = [\![48, 3, 3]\!]$, with $(d_X, d_Z) = (16, 3)$. A layer of $26\,T + 22\,T^\dagger$ implements logical CCZ with no Clifford correction.

- $\mathcal{C}_2$ is fixed by projective geometry: the columns of S are the points of $\mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2)$, the nonzero vectors of $\mathbb{F}_2^6$ outside a 4-dimensional subspace.
- The logical rows K come from a SAT (Boolean satisfiability) search.
- Lower bound: no code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$. Lengths 39 to 46 are excluded by [WJS26].
- Removing other subspaces gives codes at 13 block lengths from 48 to 496.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).

Fix S from finite projective geometry

1. Columns of S distinct and nonzero: $d_Z \geq 3$.

Fix S from finite projective geometry

1. Columns of S distinct and nonzero: $d_Z \geq 3$.

2. $\mathcal{C}_2$ is *8-divisible*: every codeword has weight $\equiv 0 \pmod 8$. By the Bravyi–Haah parity identity, pair overlaps are $\equiv 0 \pmod 4$ and triple overlaps are even, so S1–S3 hold.

Fix S from finite projective geometry

1. Columns of S distinct and nonzero: $d_Z \geq 3$.
2. $\mathcal{C}_2$ is *8-divisible*: every codeword has weight $\equiv 0 \pmod 8$. By the Bravyi–Haah parity identity, pair overlaps are $\equiv 0 \pmod 4$ and triple overlaps are even, so S1–S3 hold.
3. Search for logical rows K satisfying K1–K3 and KS1–KS3.

Fix S from finite projective geometry

1. Columns of S distinct and nonzero: $d_Z \geq 3$.
2. $\mathcal{C}_2$ is *8-divisible*: every codeword has weight $\equiv 0 \pmod 8$. By the Bravyi–Haah parity identity, pair overlaps are $\equiv 0 \pmod 4$ and triple overlaps are even, so S1–S3 hold.
3. Search for logical rows K satisfying K1–K3 and KS1–KS3.
4. Solve for the signs Γ over $\mathbb{Z}_4$.

Projectivity rules out weight-one and weight-two Z logicals

Let β_j be column j of S. A Z error on qubit j has syndrome β_j, and Z errors on qubits $i \neq j$ have syndrome $\beta_i \oplus \beta_j$.

- $\beta_j \neq 0$ for all j: every single Z error is detected.
- $\beta_i \neq \beta_j$ for all $i \neq j$: every pair of Z errors is detected.

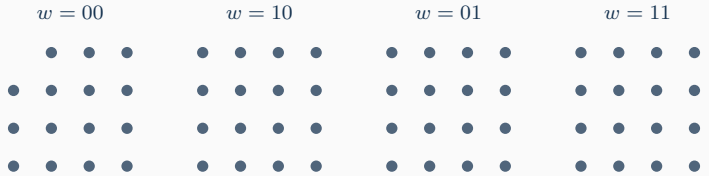
Such S is called *projective*: its columns are distinct points of $\mathrm{PG}(s-1, 2)$, the $2^s - 1$ nonzero vectors of $\mathbb{F}_2^s$.

$d_Z \geq 3$ follows from the columns of S alone, for every choice of logical rows K.

Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

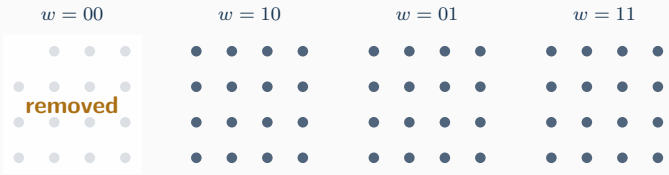
The *support* of S is its set of columns. Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$. The 48 remaining points form three blocks, one for each $w \neq 00$.



Complement of a projective subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

The *support* of S is its set of columns. Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$. The 48 remaining points form three blocks, one for each $w \neq 00$.

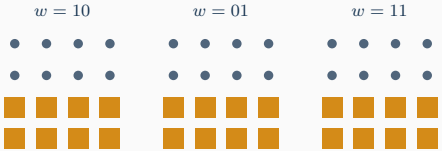


The 6 coordinate functions $u_1, \ldots, u_4, w_1, w_2$ are the rows of S.

Three blocks of 16 points give a projective $[48, 6]$ code $\mathcal{C}_2$.

Hyperplane sections and codeword weights

Each codeword c of $\mathcal{C}_2$ is a linear function $c(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on $\mathcal{P}$, and $|c| = 48 - |\mathcal{P} \cap \{c = 0\}|$.



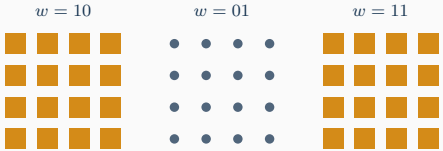
$\lambda \neq 0$, for example $c = u_1$: eight points in each block, weight $3 \cdot 8 = 24$.

Projective *two-weight* codes (two nonzero weights) correspond to point sets that meet every hyperplane in one of two sizes [CK86].

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

Hyperplane sections and codeword weights

Each codeword c of $\mathcal{C}_2$ is a linear function $c(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on $\mathcal{P}$, and $|c| = 48 - |\mathcal{P} \cap \{c = 0\}|$.



$\lambda = 0$, for example $c = w_1$: two full blocks, weight $2 \cdot 16 = 32$.

Projective *two-weight* codes (two nonzero weights) correspond to point sets that meet every hyperplane in one of two sizes [CK86].

Weights 24 and 32 are multiples of 8, so S1–S3 hold.

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

More geometric constructions

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q-1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

$p\backslash q$	1	2	3	4	5
4	none	$[\![48,3,3]\!]$, 2*	$[\![112,3,3]\!]$, 12	$[\![240,3,3]\!]$, 17	$[\![496,3,3]\!]$, 18
5	none	$[\![96,3,3]\!]$, 11	$[\![224,3,3]\!]$, 8	$[\![480,3,3]\!]$, 14	
6	$[\![64,3,4]\!]$, 5	$[\![192,3,3]\!]$, 14	$[\![448,3,3]\!]$, 13		
7	$[\![128,3,4]\!]$, 3	$[\![384,3,3]\!]$, 18			
8	$[\![256,3,4]\!]$, 3				

Each cell: parameters with $d = d_Z < d_X$, and the number of inequivalent codes found. Counts are lower bounds except $\star$ (complete). "None": excluded by our lower bound. Supports with $p \leq 3$ admit no $T/T^\dagger$ implementation of CCZ.

Outline

1. Quantum error correction and transversal gates
2. The cube code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. **Column indicators, logical rows, and phase signs**
6. Lower bounds and open questions

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. We write χ_M as a multilinear polynomial over $\mathbb{F}_2$. Its weight $|\chi_M|$ is the number of columns of M.

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. We write χ_M as a multilinear polynomial over $\mathbb{F}_2$. Its weight $|\chi_M|$ is the number of columns of M.

For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, with x for the rows of K and y for the row of S, the columns are exactly the points with $y = 1$, so $\chi_G(x, y) = y$.

Column indicators from the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. We write χ_M as a multilinear polynomial over $\mathbb{F}_2$. Its weight $|\chi_M|$ is the number of columns of M.

For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$, $u \in \mathbb{F}_2^3$:

$$G = \begin{bmatrix} 0&0&0&0&1&1&1&1 \\ 0&0&1&1&0&0&1&1 \\ 0&1&0&1&0&1&0&1 \\ 1&1&1&1&1&1&1&1 \end{bmatrix}.$$

In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$, with x for the rows of K and y for the row of S, the columns are exactly the points with $y = 1$, so $\chi_G(x, y) = y$.

$\mathcal{Q}_{48}$: the columns of S are the points with $w \neq 00$, so

$$\chi_S(u, w) = 1 + (1 + w_1)(1 + w_2) = w_1 + w_2 + w_1 w_2.$$

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g \rangle$ be the dot product of their value vectors [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t \in T} z_t$ and

$$\langle \chi_G, m_T \rangle = \left| \bigwedge_{t \in T} g_t \right| \bmod 2.$$

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g \rangle$ be the dot product of their value vectors [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t \in T} z_t$ and

$$\langle \chi_G, m_T \rangle = \left| \bigwedge_{t \in T} g_t \right| \bmod 2.$$

The CH conditions say: $\langle \chi_G, x_1 x_2 x_3 \rangle = 1$, and $\langle \chi_G, m \rangle = 0$ for every other monomial of degree 1 to 3.

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g\rangle$ be the dot product of their value vectors [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t\in T} z_t$ and

$$\langle \chi_G, m_T\rangle = \Big|\bigwedge_{t\in T} g_t\Big| \bmod 2.$$

The CH conditions say: $\langle \chi_G, x_1x_2x_3\rangle = 1$, and $\langle \chi_G, m\rangle = 0$ for every other monomial of degree 1 to 3.

Since $\mathrm{RM}(3,r)^\perp = \mathrm{RM}(r-4,r)$, and $y_1\cdots y_s$ (one at the 8 points $(x,1,\dots,1)$) satisfies these conditions [LB26],

$$\chi_G \in y_1\cdots y_s + \varepsilon\delta_0 + \mathrm{RM}(r-4,r), \qquad r = 3+s,\ \varepsilon = n \bmod 2,$$

where δ_0 is the indicator of the origin.

Logical rows in the column picture

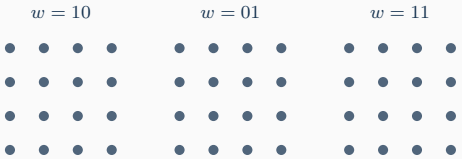
Column $\beta = (u, w) \in \mathcal{P}$ of S has the *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the corresponding column of K [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

Split a set T of rows into its logical rows T_K and stabilizer rows T_S. The CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{T_K}\big(\kappa(\beta)\big)\, m_{T_S}(\beta) = [T = \{K_1, K_2, K_3\}].$$

S1–S3 have $T_K = \emptyset$ and hold because $\mathcal{C}_2$ is 8-divisible. The rest have degree at most 3 in the label bits.



Logical rows in the column picture

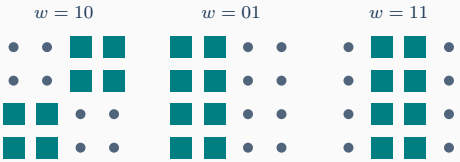
Column $\beta = (u, w) \in \mathcal{P}$ of S has the *logical label* $\kappa(\beta) \in \mathbb{F}_2^3$, the corresponding column of K [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

Split a set T of rows into its logical rows T_K and stabilizer rows T_S. The CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{T_K}\big(\kappa(\beta)\big)\, m_{T_S}(\beta) = [T = \{K_1, K_2, K_3\}].$$

S1–S3 have $T_K = \emptyset$ and hold because $\mathcal{C}_2$ is 8-divisible. The rest have degree at most 3 in the label bits.



Teal: $\kappa_1 = 1$, with $\kappa_1 = u_1 + u_3$, $1 + u_3$, u_4 on the three blocks.

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) g(\beta) \bmod 2$ [LB26].

1. Let y_l be the l-th coordinate on $\mathcal{P}$ (row l of S). K1, KS1, and KS3 for one row mean $\kappa_i \perp \mathbf{1}, y_l, y_l y_{l'}$ on $\mathcal{P}$. These span $\mathrm{RM}(2, s)$, so κ_i lies in $V_{\mathcal{P}}$, the orthogonal complement of $\mathrm{RM}(2, s)$ on $\mathcal{P}$, a linear condition.

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g \rangle_{\mathcal{P}} = \sum_{\beta \in \mathcal{P}} f(\beta) g(\beta) \bmod 2$ [LB26].

1. Let y_l be the l-th coordinate on $\mathcal{P}$ (row l of S). K1, KS1, and KS3 for one row mean $\kappa_i \perp \mathbf{1}, y_l, y_l y_{l'}$ on $\mathcal{P}$. These span $\mathrm{RM}(2, s)$, so κ_i lies in $V_{\mathcal{P}}$, the orthogonal complement of $\mathrm{RM}(2, s)$ on $\mathcal{P}$, a linear condition.
2. Choose a pair $\kappa_1, \kappa_2 \in V_{\mathcal{P}}$ with $\langle \kappa_1 \kappa_2, \mathbf{1} \rangle_{\mathcal{P}} = \langle \kappa_1 \kappa_2, y_l \rangle_{\mathcal{P}} = 0$ (K2 and KS2).

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g\rangle_{\mathcal{P}} = \sum_{\beta\in\mathcal{P}} f(\beta)g(\beta) \bmod 2$ [LB26].

1. Let y_l be the l-th coordinate on $\mathcal{P}$ (row l of S). K1, KS1, and KS3 for one row mean $\kappa_i \perp \mathbf{1}, y_l, y_l y_{l'}$ on $\mathcal{P}$. These span $\mathrm{RM}(2, s)$, so κ_i lies in $V_{\mathcal{P}}$, the orthogonal complement of $\mathrm{RM}(2, s)$ on $\mathcal{P}$, a linear condition.
2. Choose a pair $\kappa_1, \kappa_2 \in V_{\mathcal{P}}$ with $\langle \kappa_1\kappa_2, \mathbf{1}\rangle_{\mathcal{P}} = \langle \kappa_1\kappa_2, y_l\rangle_{\mathcal{P}} = 0$ (K2 and KS2).
3. For a fixed pair, the remaining conditions on $\kappa_3 \in V_{\mathcal{P}}$, namely $\langle \kappa_3, \kappa_i\rangle_{\mathcal{P}} = \langle \kappa_3, y_l\kappa_i\rangle_{\mathcal{P}} = 0$ and $\langle \kappa_3, \kappa_1\kappa_2\rangle_{\mathcal{P}} = 1$, are linear. Gaussian elimination over $\mathbb{F}_2$ finds κ_3 or shows that none exists.

Finding the logical rows

Fix $\mathcal{P}$. Logical row i is $\kappa_i \in \mathbb{F}_2^{\mathcal{P}}$, and $\langle f, g\rangle_{\mathcal{P}} = \sum_{\beta\in\mathcal{P}} f(\beta)g(\beta) \bmod 2$ [LB26].

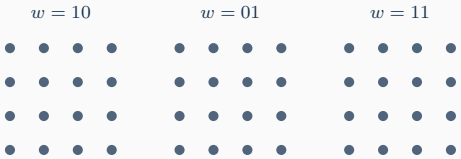
1. Let y_l be the l-th coordinate on $\mathcal{P}$ (row l of S). K1, KS1, and KS3 for one row mean $\kappa_i \perp \mathbf{1}, y_l, y_l y_{l'}$ on $\mathcal{P}$. These span $\mathrm{RM}(2, s)$, so κ_i lies in $V_{\mathcal{P}}$, the orthogonal complement of $\mathrm{RM}(2, s)$ on $\mathcal{P}$, a linear condition.
2. Choose a pair $\kappa_1, \kappa_2 \in V_{\mathcal{P}}$ with $\langle \kappa_1\kappa_2, \mathbf{1}\rangle_{\mathcal{P}} = \langle \kappa_1\kappa_2, y_l\rangle_{\mathcal{P}} = 0$ (K2 and KS2).
3. For a fixed pair, the remaining conditions on $\kappa_3 \in V_{\mathcal{P}}$, namely $\langle \kappa_3, \kappa_i\rangle_{\mathcal{P}} = \langle \kappa_3, y_l\kappa_i\rangle_{\mathcal{P}} = 0$ and $\langle \kappa_3, \kappa_1\kappa_2\rangle_{\mathcal{P}} = 1$, are linear. Gaussian elimination over $\mathbb{F}_2$ finds κ_3 or shows that none exists.

If no pair extends, $\mathcal{P}$ is excluded. A SAT solver can also solve the full system at once. For $\mathcal{Q}_{48}$, the columns at $u = 0000$ in the three blocks sum to 0 and their labels to $011 \oplus 110 \oplus 000 = 101$. So Z on these three qubits is a logical operator, and $d_Z = 3$.

Solving for the $T/T^\dagger$ pattern

Write $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j \in \{0, 1\}$. Then $\mathrm{sw}_\Gamma(c) = |c| - 2|\gamma \wedge c|$, and every codeword $c = K^\top x \oplus S^\top y$ has even weight:

$$|\gamma \wedge c| \equiv \tfrac{1}{2}|c| - 2x_1x_2x_3 \pmod 4.$$



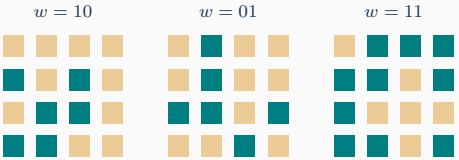
Solving for the $T/T^\dagger$ pattern

Write $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j \in \{0, 1\}$. Then $\mathrm{sw}_\Gamma(c) = |c| - 2|\gamma \wedge c|$, and every codeword $c = K^\top x \oplus S^\top y$ has even weight:

$$|\gamma \wedge c| \equiv \tfrac{1}{2}|c| - 2x_1x_2x_3 \pmod 4.$$

For $\mathcal{Q}_{48}$ [LB26]:

$$\begin{aligned}\gamma(u, w) &= u_3 + w_1 Q_1(u) + w_2 Q_2(u),\\ Q_1 &= u_1 + u_2 + u_3 + u_1u_3 + u_2u_3 + u_2u_4,\\ Q_2 &= u_3 + u_4 + u_1u_2 + u_1u_4 + u_3u_4.\end{aligned}$$



Teal: $T^\dagger$ $(6, 6, 10$ per block). Gold: T. Total $26\,T + 22\,T^\dagger$.

Outline

1. Quantum error correction and transversal gates
2. The cube code: a worked example of logical CCZ
3. Triorthogonality and the CCZ conditions
4. Projective geometry and the 48-qubit code
5. Column indicators, logical rows, and phase signs
6. **Lower bounds and open questions**

Stabilizer supports are Reed–Muller codewords

For projective S, S1–S3 say $\langle \chi_S, m \rangle = 0$ for every monomial m in $y_1, \ldots, y_s$ of degree 1 to 3.

Following [NH22], let $c = \chi_S + \varepsilon \delta_0$, where ε makes $|c|$ even. Every monomial of degree at least 1 vanishes at 0, so adding δ_0 changes only $\langle c, \mathbf{1} \rangle = |c| \bmod 2$. Hence $c \perp \mathrm{RM}(3, s)$ and

$$c \in \mathrm{RM}(3, s)^{\perp} = \mathrm{RM}(s - 4, s), \qquad n = |c| - c(0).$$

For $\mathcal{Q}_{48}$: $c = \chi_S = w_1 + w_2 + w_1 w_2 \in \mathrm{RM}(2, 6)$.

Our exclusion argument covers every length through 38

1. Reduce the stabilizer rank s (the number of rows of S) to $s \leq 8$ when $n \leq 38$.
2. Exclude $s \leq 5$, then settle ranks 6 and 7 using the possible weights and the geometry of the support.
3. At rank 8, use the classification of low-weight Reed–Muller codewords to reduce to finitely many supports, and enumerate them.
4. Show that no remaining support admits logical rows satisfying K1–K3 and KS1–KS3.

Independent result of our paper

No CSS code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$.

Existence of a valid code at different block lengths

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	[WJS26]
$n = 47$	Quasi-transversal CCZ	[JVB+26]
$n = 48$	Physical $T/T^\dagger$, no correction	Our explicit code

Wills–Jain–Singh classify generalized triorthogonal codes with $n \leq 54$ and $d_Z \geq 3$, which covers quasi-transversal CCZ with $k = 3$.

The $[\![47, 3, 3]\!]$ code of [JVB+26] has no physical $T/T^\dagger$ layer [LB26]. Its stabilizer support is $\mathbb{F}_2^6$ minus an affine 4-flat (a translate of a 4-dimensional subspace), with the origin deleted. $\mathcal{Q}_{48}$ removes a linear 4-dimensional subspace instead. [JVB+26] report a 47-to-1 CCZ distillation protocol from it with a circuit footprint of 9 qubits.

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).

Takeaways and open questions

Takeaways

- Projective geometry fixes S: $d_Z \geq 3$ and weights divisible by 8.
- Column indicators turn the CH conditions into Boolean equations. The signs solve a $\mathbb{Z}_4$ system.
- $\mathcal{Q}_{48} = [\![48, 3, 3]\!]$ with $26\,T + 22\,T^\dagger$. Our bound excludes $n \leq 38$, and [WJS26] excludes 39–46.

Open questions

- Which supports admit logical rows K and signs Γ?
- Higher distance, other gates, more logical qubits.
- Minimum length for $T/T^\dagger$ CCZ with no correction.
- Full circuits, noise, and resource costs.

Paper
arXiv:2610.09341

Code and data
github.com/jerrylvx/gate2code-public